卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0308 30-B003701S-T0308

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 03 卷 第 08 册
Structure SPECIALITY VOLUME 03 BOOK 08

卷册 名称 直流中性母线塔加工图
Volume name Manufacture for DC Neutral Bus-bar Tower

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	古韬城
图纸 5 张 DRAWING 5 NOS.	打印件 0 本 PRINTED COPIES 0NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0308-001	钢结构加工说明 Specification for Fabrication and Design of Steel Works	A	1		
2	30-B003701S-T0308-002	直流中性母线塔布置图 Layout of DC Neutral Bus-bar Tower	A	1		
3	30-B003701S-T0308-003	直流中性母线塔身结构图 ① Superstructure for DC Neutral Bus-bar Tower ①	A	1		
4	30-B003701S-T0308-004	直流中性母线塔身结构图 ② Superstructure for DC Neutral Bus-bar Tower ②	A	1		
5	30-B003701S-T0308-005	直流中性母线塔腿结构图 ③ Superstructure for DC Neutral Bus-bar Tower ③	A	1		

注:

SPECIFICATION FOR FABRICATION AND DESIGN OF STEEL WORKS

1. GENERAL NOTES:

All dimensions are in millimeters unless otherwise noted.

The elevations are in meters.

All steel members should adopt the anti-corrosive treatment of hot dip galvanizing,the galvanized standard shall conform to the latest revision of ASTM-A123M & ASTM-A153M.

2.MATERIALS:

Structural steel:

The grade of steel used in the structure shall be Q345B(Fy=345Mpa ,Fu=470Mpa) or Q235B(Fy=235Mpa ,Fu=375Mpa).

The steel shall confirm to《GB/T 1591-2008 High strength low alloy structural steels》

Structural bolts, nuts and washers:

Bolts shall be of steel grade 6.8 or grade 8.8, hot dip galvanized, with match nuts and washers;

Nuts and washers shall be of steel grade 6.8 or grade 8.8, hot dip galvanized.

The bolts and nuts shall confirm to《DL/T 284-2012 Hot-dip galvanized bolts and nuts for tower and hardware of transmission lines》

If there is no special description,the connection use single nut, includes one standard nut, one plain washer, one spring washer;

And if there is special description,the connection use Double nut, includes one standard nut, one thin nut, one plain washer.

Welding rod:

The type of welding rod shall match with the steel.

For Q345B steel, the type is E50xx(F_{E50}=490Mpa);

For Q235B steel, the type is E43xx(F_{E43}=420Mpa);

3.MANUFACTURE:

Only after lofting, all members and gusset plates can be cut in the manufactory. The sizes of members listed in the steel schedule are only used for the preparation of steel.

During the fabrication process of steel gantry tower and girder members, all members shall be pre-assembled before producing in batches.

Only after steel members of gantry tower and girder must match the requirements of the relevant design drawings and standards, they can be transported to the construction-site.

Welding shall be finished before hot dip galvanizing.

In order to avoid bending, the hanging points shall be properly selected while galvanized steel members are taken out from the galvanization caldron,

All production welding shall be carried out in accordance with the requirements of welding.

The leg size of fillet welds shoud not be less than the thickness of the thinner part welded, but not more than 1.2 times that.

For the thick plate , grooving should be adopted in order to ensure the strength of connection.

The spacing between ends of angles shall be 10mm.

For the type of inside splice angle, the angle of splice angle will be filleted.

For the type of outer splice angle, the inside arc of splice angle will be cut into right-angle.

The standard dimensions of steel angle neat line and bolt, max.allowable diameter of bolt hole should be taken from Table 1, where not otherwise noted in the drawing .

The standard spacing between centres of bolts , edge and end distances on steel angles and gussets should be taken from Table 2, where not otherwise noted in the drawing.

the component shall adopt the technics to stop the lamellar tearing if the thickness of fillet weld is more than 20mm.

The bolt at the column feet adopts double nuts.

4.TRANSPORTATION AND STACKING:

During the transportation, the same kind of members should be tied up into bundles. To avoid the deformation of the members and the damage of galvanized layer due to mutual collision, the sleepers should be lined at the bottom of the members and between their layers.

After the transporting the membes to the construction-site, they should be piled separately according to the numbe of gantry tower and girder to avoid confusion during assembly.

The damaged part of zinc layer, which is caused by transportation and hoisting, should be cleaned and then brushed with zinc rich paint.

5.After erection of gantry,the holding-down pad shall be welded with the leg slab ,then brushed with zinc rich paint.

6.All steels should be rolled steel and shall confirm to local standards.

7.welding

1).the welding Procedures and welder-certification programs should be in accordance with the latest revision of the American welding society's Structural Welding Code for Steel (AWS D1 1/D1 1M).

1. 总说明：

除特殊注明外，所有标注尺寸单位为毫米。

标高标注的单位为米。

所有钢构件采用热浸锌防腐处理。热镀锌标准应符合最新版规范ASTM-A123M以及ASTM-A153规定。

2. 材料：

钢材：

钢结构所用的钢材为Q345B(Fy=345Mpa ,Fu=470Mpa)或Q235B(Fy=235Mpa ,Fu=375Mpa).

钢材需要满足《低合金高强度结构钢(GB/T 1591-2008)》的要求。

螺栓、螺母和垫圈：

螺栓应为6.8或 8.8级的热浸锌件，带有螺母、平垫圈；

螺母、垫圈应为相应的热浸锌件。

螺栓螺母需满足《 输电线路杆塔及电力金具用热浸镀锌螺栓与螺母(DL/T 284-2012) 》要求。

图纸中连接无特殊注明处，螺栓均为单帽，：一标准帽一平一弹。

图纸中连接注明为双帽的，螺栓均为双帽，：一标准帽一薄帽一平。

焊条：

焊条类型应与连接母材材质匹配。

对于Q345钢材，采用E50xx型焊条(F_{E50}=490Mpa).

对于Q235钢材，采用E43xx型焊条(F_{E43}=420Mpa).

3. 加工制造：

所有构件及节点板均需放样后下料，钢材明细表中的构件尺寸仅供备料用。

钢结构加工过程中必须进行试组装，然后才能成批加工。

钢结构加工必须满足相关图纸和验收规范要求方能运输到现场。

焊接构件应在完成焊接后再镀锌。

为避免镀锌构件从锌锅中取出时发生弯曲，应选择合适的吊点。

构件焊接应按照相应的焊接规程、规范的要求进行。

角焊缝高度不小于连接构件的最小厚度，但不大于1.2倍的连接构件的最小厚度。

对于较厚的焊件需采用坡口焊，保证连接强度。

两角钢对接接头处的间隙为10mm。

内包角钢连接时，采用铲内包角钢背的方法，外包角钢时采用铲外包角钢弧的方法。

角钢基准线和螺栓准线除图中注明外，应按表1采用。

角钢及节点板的螺栓间距、边距除图中注明外，应按表2采用。

柱脚锚栓采用双螺母。

4. 运输与堆放：

运输期间，同类构件应捆绑成束堆放在一起。枕木放在构件末端和层间以防止变形和由于相互碰撞带来镀锌层的破坏。

构件运输到现场后，柱架构件应分离堆放，以防组装期间带来混淆。

由于运输和吊装破坏的镀锌层部分，应先清洗，然后涂刷富锌漆。

5. 构架安装完毕后，地脚螺栓垫板与柱脚焊接，刷富锌漆防腐。

6. 所有钢材均为轧制，且应符合当地标准。

7.焊接

1).焊接工艺和焊工资格认证程序需满足现行美国焊接协会《钢结构焊接规范AWS D1 1M》的要求。

TABLE 1

No. 编号	FLANGE DIMENSION 角钢肢宽 b	NEAT LINE DIMENSION 基准规线距 a0	BOLT DIMENSION 螺栓准线距 SINGLE LINE 单排 a0	DOUBLE LINES 双排 a1	DOUBLE LINES 双排 a2	MAX. ALLOWABLE DIAMETER OF BOLT HOLE 最大允许开孔直径 ø
1	50	25	25			17.5
2	56	28	28			
3	63	30	30			
4	70	35	35			21.5
5	75	38	38			
6	80	40	40			
7	90	45	45			
8	100	50	50			
9	110	55	55	45	75	
10	125	60	60	50	85	
11	140	70	70	55	90	25.5
12	160	80	80	60	105	
13	180	90	90	65	120	
14	200	100	100	75	135	

Note: The unit of all values listed in the table is millimeter.
说明： 表格中所有数值均以毫米为单位。

TABLE 2

NOMINAL DIA. OF BOLT 螺栓直径	DIA. OF HOLE 螺栓孔直径	DISTANCE BETWEEN CENTRES 螺栓孔距 SINGLE LINE 单排	DISTANCE BETWEEN CENTRES 螺栓孔距 DOUBLE LINES 双排	EDGE AND END DISTANCE 螺栓边距及端距 Ld	Lz	La
16	17.5	50	80	25	≮21	≮23
20	21.5	60	100	30	≮26	≮28
24	25.5	80	120	40	≮31	≮33

Note: 1.the unit of all values listed in the table is millimeter.
说明： 1.表格中所有数值均以毫米为单位。
2."Lz" is the edge distance for a rolled edge.
2."Lz" 表示轧制边距。
3."La" is the edge distance for a sheared edge.
3."La" 表示切角边距。
4."Ld" is the end distance.
4."Ld"表示端距。

BILLS OF BOLT TYPE
螺栓类型明细表

TYPE 类型	THROUGH THICK 通过厚度
M16X40	8~12
M16X50	13~22
M16X60	23~32
M20X45	10~15
M20X55	16~25
M20X65	26~35
M20X75	36~45
M20X85	46~55
M20X95	56~65
M24X55	16~20
M24X65	21~30
M24X75	31~40

PRETIGHTENING RORQUE VALUE OF BOLT FOR GRADE 6.8
6.8螺栓预紧扭矩值(N.m)

DIAMETER 螺栓直径	PRETIGHTENING RORQUE VALUE 预紧扭矩
M12	40
M16	80
M20	160
M24	280
M27	360

PRETIGHTENING RORQUE VALUE OF BOLT FOR GRADE 8.8
8.8螺栓预紧扭矩值(N.m)

DIAMETER 螺栓直径	PRETIGHTENING RORQUE VALUE 预紧扭矩
M16	110
M20	220
M24	380
M27	450
M30	600
M33	700
M36	880
M39	1100
M42	1400
M45	1900

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 8 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	古韬 喊	2018.12.13
校核 Checked	王惠平	2018.12.13
审核 Reviewed	张清晴	2018.12.13
批准 Approved	杨军	2018.12.13

钢结构加工说明

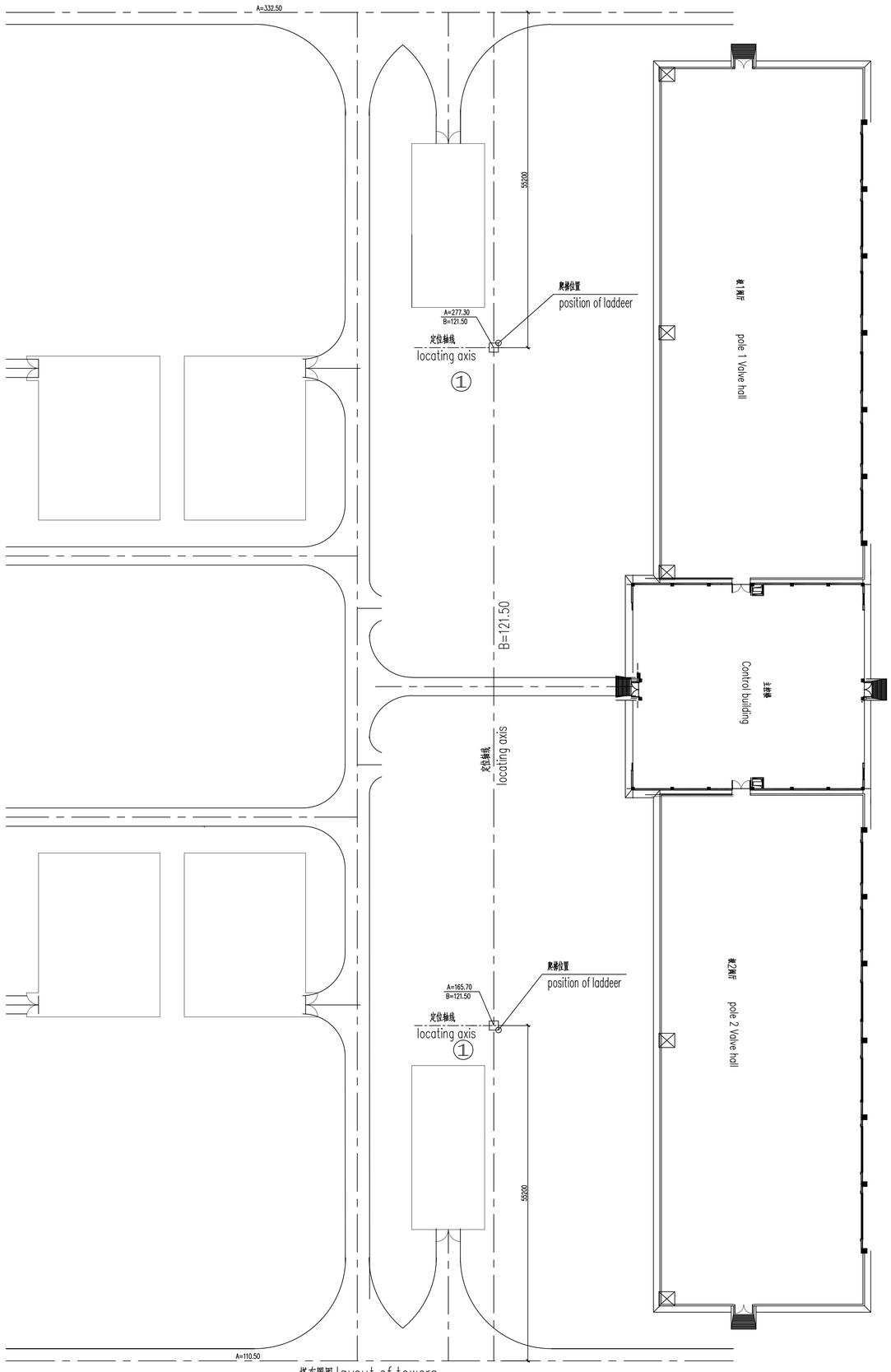
Specification For Fabrication And Design Of Steel Works

比例 Scale: 1:100

图号 DWG NO.: 30-B003701S-T0308-001

版本 REV.: A

420X594



塔布置图 layout of towers

北 N

LISTS OF steelworks 钢结构工程量

NO 编号	QTY 数量	DRAWING NO. 施工图号	mass工程量(kg) PCS单个	SUM总计
①	2	30-B003701S-T0308-003~005	2067.1	4134.20

说明:

1. 场地±0.000相当于绝对标高193.40(EGM2008高程系统).如无特别说明,本工程标高均采用相对标高.
2. 本工程初设设计尚未最终审定,最终方案有存在调整可能,是否按图施工由总包CET决定,由此造成的变更设计不予负责.

Specification:

1. The site level of ±0.000 is equal to the absolute elevation of193.40 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.
2. As the preliminary design documents have not yet been approved finally, the project solution has probability to be modified.The design contractor has no responsibility for the project solution changed, if the construction contractor does not get authorized by CET before construction.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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M L H V D C - L C S - E C I - D D - C S - 0 3 0 8 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	古韬	2018.12.13
校核 Checked	毛惠平	2018.12.13
审核 Reviewed	马清清	2018.12.13
批准 Approved	杨之	2018.12.13
比例 Scale	1:400	图号 DWG NO. 30-B003701S-T0308-002

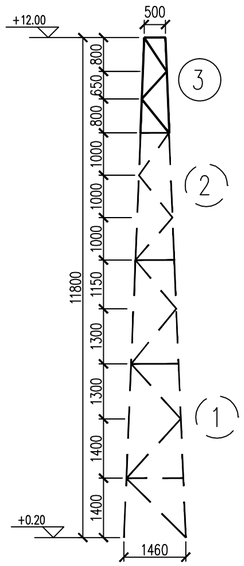
专业 Speciality	会签 Sign	日期 Date
电气	郑煜	2018.06.25

直流中性母线塔布置图

Layout of DC Neutral Bus-bar Tower

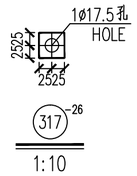
版本 REV. A

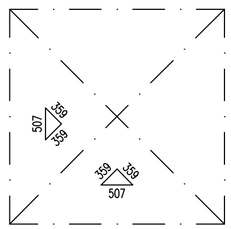
594X841



vertical view of tower of guard cable

单线图





螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备注 REMARK
螺栓 BOLT	6.8	M16X70		1	0.2	
		M20X45		24	6.5	
		M20X55		64	18.9	
		M20X65		4	1.3	
		M16X70		4	0.9	双螺母 TWO NUTS
		M20X80		32	13.2	双螺母 TWO NUTS
脚钉 STEP-BOLT	6.8	M16X180		3	1.0	
		M20X200		1	0.6	
垫圈 WASHER	Q235	-4(ø22)	规格X个数 SPECS X Qty	16	0.3	
合计 TOTAL (kg)		42.9 kg				
总计 SUM TOTAL (kg)		335.6 kg				

构件明细表
BILL OF MATERIAL

编号 NO.	规格 SPECS	长度(mm) LENGTH	数量 Qty	WEIGHT(kg) 重量 一件 PIECE	小计 SUM	备注 REMARK
301	Q345L80X8	1874	1	18.10	18.1	带脚钉 WITH STEP BOLTS
302	Q345L80X8	1874	1	18.10	18.1	
303	Q345L80X8	1874	1	18.10	18.1	
304	Q345L80X8	1874	1	18.10	18.1	
305	L63X5	1087	4	5.24	21.0	切角 CHAMFERED
306	L63X5	941	4	4.54	18.2	
307	L63X5	867	4	4.18	16.7	切角 CHAMFERED
308	L70X6	400	4	2.56	10.2	
309	-6X142	244	4	1.63	6.5	
310	-8X144	277	4	2.50	10.0	
311	-8X202	260	4	3.30	13.2	
312	-8X192	202	4	2.44	9.8	
313	L63X5	527	2	2.54	5.1	
314	L70X6	200	4	1.28	5.1	
315	L70X6	200	4	1.28	5.1	
316	Q345-26X348	348	4	24.72	98.9	
317	-26X50	50	1	0.51	0.5	垫块 CUSHION BLOCK
合计 TOTAL (kg)		292.7 kg				

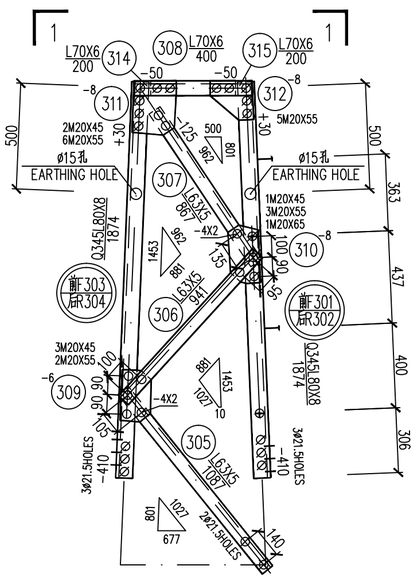
说明:

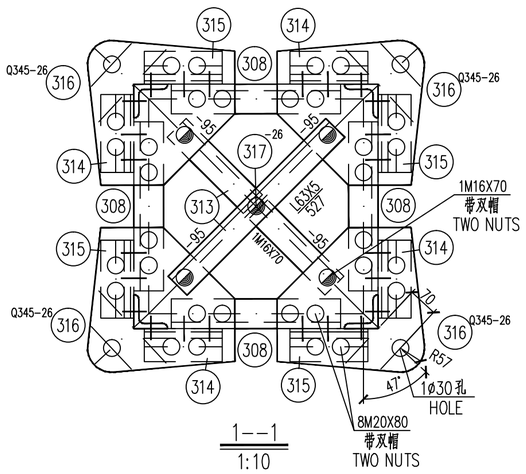
1、构件和板未注明的材质为Q235B。

2、材料表中尺寸仅供备料参考,施工放样定。

Specification:

1. The material of the component and board not specified is Q235B.
2. The dimensions in the material table are for reference only.





1--1
1:10

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

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±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 8 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	古绍斌	2018.12.13
校核 Checked	王惠平	2018.12.13
审核 Reviewed	邹清清	2018.12.13
批准 Approved	杨涛	2018.12.13
比例 Scale	1:20	图号 DWG NO.

直流中性母线身部结构图 ③

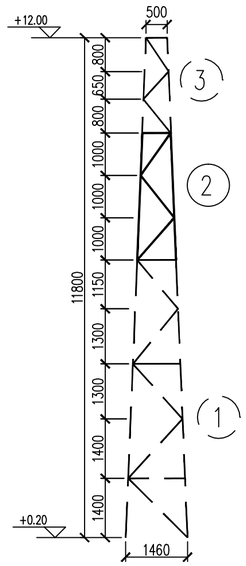
Superstructure for DC Neutral Bus-bar Tower ③

30-B003701S-T0308-005

版本 REV. A

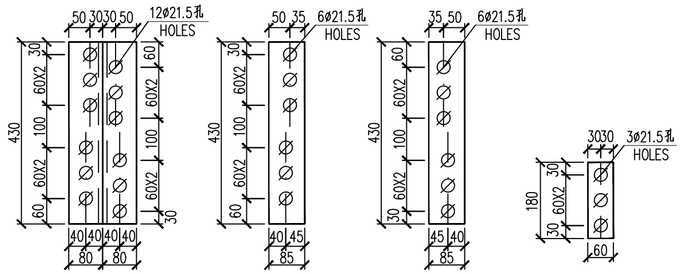
电气	郑煜	2018.06.25
专业 Speciality	会签 Sign	日期 Date

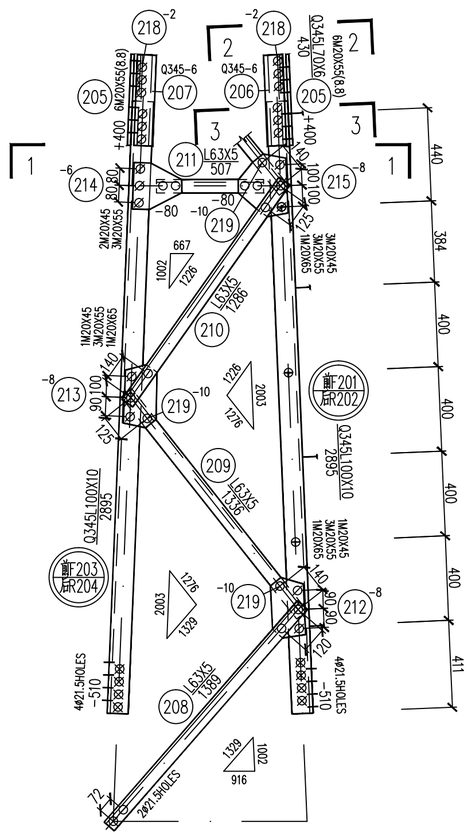
420X594

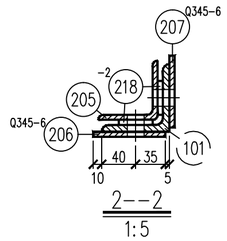


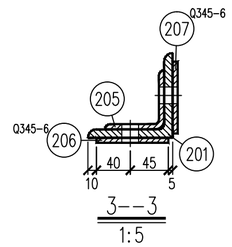
vertical view of tower of guard cable

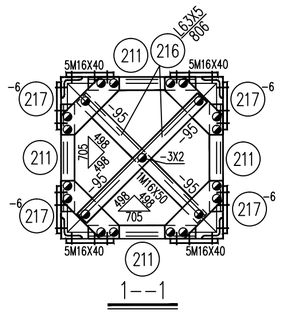
单线图

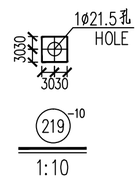












螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备注 REMARK
螺栓 BOLT	6.8	M16X40	◐	20	2.9	
		M16X50	◐	1	0.2	
		M20X45	○	28	7.6	
		M20X55	∅	48	14.2	
		M20X65	⊗	12	3.8	
	8.8	M20X55	∅	48	14.2	
脚钉 STEP-BOLT	6.8	M16X180	⊕	5	1.6	
		M20X200		2	1.2	
垫圈 WASHER	Q235	-3(ø17.5)	规格X个数 SPECS X Qty	2	0.1	
合计 TOTAL (kg)		45.8 kg				
总计 SUM TOTAL (kg)		401.8 kg				

构件明细表
BILL OF MATERIAL

编号 NO.	规格 SPECS	长度(mm) LENGTH	数量 Qty	WEIGHT(kg) 重量 一件 PIECE	小计 SUM	备注 REMARK
201	Q345L100X10	2895	1	43.77	43.8	带脚钉 WITH STEP BOLTS
202	Q345L100X10	2895	1	43.77	43.8	
203	Q345L100X10	2895	1	43.77	43.8	
204	Q345L100X10	2895	1	43.77	43.8	
205	Q345L70X6	430	4	2.75	11.0	铲背 HEEL FILLETING
206	Q345-6X85	430	4	1.72	6.9	
207	Q345-6X85	430	4	1.72	6.9	
208	L63X5	1389	4	6.70	26.8	
209	L63X5	1336	4	6.44	25.8	切角 CHAMFERED
210	L63X5	1286	4	6.20	24.8	
211	L63X5	507	4	2.44	9.8	
212	-8X157	287	4	2.83	11.3	
213	-8X154	298	4	2.88	11.5	
214	-6X232	220	4	2.40	9.6	
215	-8X232	273	4	3.98	15.9	
216	L63X5	806	2	3.89	7.8	
217	-6X108	387	4	1.97	7.9	
218	-2X60	180	8	0.17	1.4	垫块 CUSHION BLOCK
219	-10X60	60	12	0.28	3.4	垫块 CUSHION BLOCK
合计 TOTAL (kg)		356.0 kg				

说明:

1、构件和板未注明的材质为Q235B。

2、材料表中尺寸仅供备料参考,施工放样定。

Specification:

1. The material of the component and board not specified is Q235B.
2. The dimensions in the material table are for reference only.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 8 - 0 0 4 A

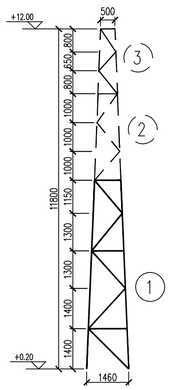
	姓名 Name	日期 Date
设计 Designed	古超城	2018.12.13
校核 Checked	王惠平	2018.12.13
审核 Reviewed	邹清晋	2018.12.13
批准 Approved	杨立平	2018.12.13

直流中性母线塔身部结构图 ②

Superstructure for DC Neutral Bus-bar Tower ②

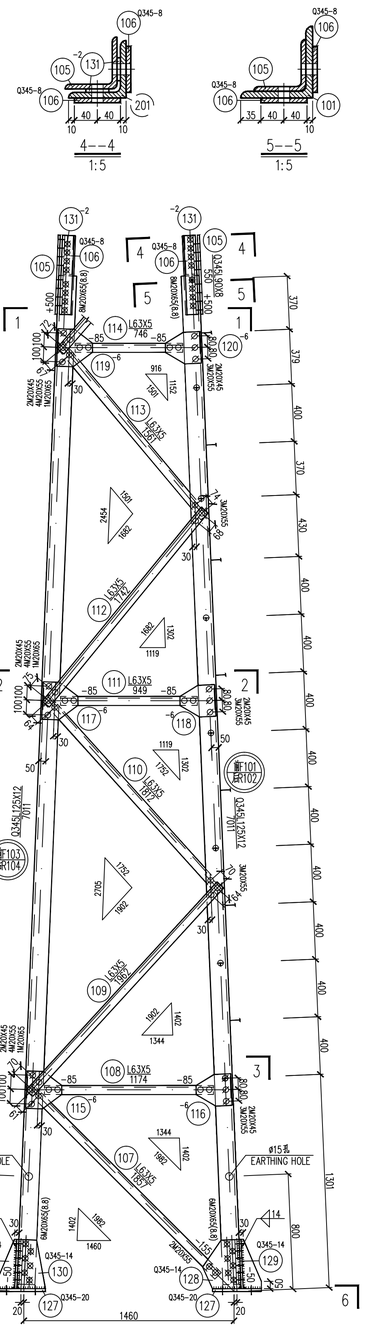
比例 Scale	1:20	图号 DWG NO.	30-B003701S-T0308-004	版本 REV.	A

420X594



vertical view of tower of guard cable

单线图



说明:

1、构件和板未注明的材质为Q235B。

2、材料表中尺寸仅供备料参考,施工放样定。

Specification:

1. The material of the component and board not specified is Q235B.
2. The dimensions in the material table are for reference only.

构 件 明 细 表
BILL OF MATERIAL

编号 NO.	规 格 SPECS	长度(mm) LENGTH	数量 Qty	WEIGHT(kg) 重量 一件 PIECE	合计 SUM	备 注 REMARK
101	Q345L125X12	7011	1	159.12	159.1	带脚钉 WITH STEP BOLTS
102	Q345L125X12	7011	1	159.12	159.1	
103	Q345L125X12	7011	1	159.12	159.1	
104	Q345L125X12	7011	1	159.12	159.1	
105	Q345L90X8	550	4	6.02	24.1	铲背 HEEL FILLETING
106	Q345-8X80	550	8	2.76	22.1	
107	L63X5	1857	4	8.95	35.8	切角 CHAMFERED
108	L63X5	1174	4	5.66	22.6	
109	L63X5	1962	4	9.46	37.8	
110	L63X5	1812	4	8.74	35.0	切角 CHAMFERED
111	L63X5	949	4	4.58	18.3	
112	L63X5	1742	4	8.40	33.6	
113	L63X5	1561	4	7.53	30.1	切角 CHAMFERED
114	L63X5	746	4	3.60	14.4	
115	-6X247	260	4	3.02	12.1	
116	-6X220	247	4	2.56	10.2	
117	-6X247	260	4	3.02	12.1	
118	-6X220	247	4	2.56	10.2	
119	-6X247	260	4	3.02	12.1	
120	-6X220	247	4	2.56	10.2	
121	L63X5	1119	2	5.40	10.8	
122	-6X108	394	4	2.00	8.0	
123	L63X5	1406	2	6.78	13.6	
124	-6X108	395	4	2.01	8.0	
125	L63X5	1723	2	8.31	16.6	
126	-6X108	394	4	2.00	8.0	
127	Q345-20X340	340	4	18.15	72.6	电焊 WELDING
128	Q345-14X343	408	4	15.38	61.5	电焊 WELDING
129	Q345-14X129	348	4	4.93	19.7	电焊 WELDING
130	Q345-14X199	342	4	7.48	29.9	电焊 WELDING
131	-2X60	240	8	0.23	1.8	垫块 CUSHION BLOCK
合计 TOTAL (kg)		1227.6 kg				

螺栓.脚钉.垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规 格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备 注 REMARK
螺栓 BOLT	6.8	M16X40		60	8.7	
		M16X50		3	0.5	
		M20X45		48	13.0	
		M20X55		116	34.3	
		M20X65		12	3.8	
	8.8	M20X65		112	35.9	
脚钉 STEP-BOLT	6.8	M16X180		12	3.9	
		M20X200		3	1.9	
垫圈 WASHER	Q235	-3(ø17.5)	规格X个数 SPECS X Qty	6	0.1	
合计 TOTAL (kg)		102.1 kg				
总计 SUM TOTAL (kg)		1329.7 kg				

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

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设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 8 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	古韬	2018.12.13
校核 Checked	王惠平	2018.12.13
审核 Reviewed	邹浩清	2018.12.13
批准 Approved	韩和平	2018.12.13
比例 Scale	1:20	图号 DWG NO.

直流中性母线塔腿身结构图 ①

Extension Leg for DC Neutral Bus-bar Tower ①

30-B003701S-T0308-003

版本 REV. A

电气 Speciality	郑煜 Sign	2018.06.25 Date
专业	会签	日期

594X841